



NITI Aayog

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POLICY AT A GLANCE

Electronics

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Electronics

Manufacturing in India

The **'Policy-at-a-Glance'** provides users with curated information for first-level analysis and inter-state comparisons of policy. The document brings together all topic-relevant national and state-level acts, policies, and schemes in one place. Further, it identifies for a reader what policy features are covered in which document in a simple table.

The first section of this document sets out some 'Context and Background' on the topic; the subsequent sections explain the 'Policy Features' that have been compared and include a 'Glossary' of technical terms used; the last section is the 'Table of Insights' that marks out what 'Policy Feature' has been covered in each of the topic-relevant documents

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1. Context

Scope and Objective:

The Indian electronics industry is the world's largest and fastest-growing industry, increasingly finding applications in all sectors of the economy. The domestic production of electronic items increased from Rs 3,17,331 crore (USD 49 billion) in fiscal 2017 to Rs 6,40,810 crore (USD 87.1 billion) in fiscal 2022, logging a compound annual growth rate (CAGR) of 15%. According to the Ministry of Electronics and Information Technology, India's electronics production is expected to reach USD 300 billion by 2026. To meet the demand, the central government has released niche policies/schemes over the National Policy on Electronics 2019, such as the Production Linked Incentive (PLI) scheme for phones, IT hardware and electronic components, as well as the India Semiconductor Mission. In line with the Centre, the state governments have also sharpened their focus on promoting electronics manufacturing. These schemes have been successful in attracting global champions, while providing a fillip to domestic companies and fostering their global aspirations.

2. Policy features

The section identifies a common set of policy features that documents referred to in Section 4 (Table of Insights) cover. The features are classified under the following five broad aspects

- A. Exports
- B. Employment
- C. Financial incentives
- D. Ease of doing business
- E. Non fiscal support

Dedicated electronics policy: With the increasing demand for electronics, India is taking centre stage, aiming to both meet its domestic needs and support global demand. Having a dedicated electronics policy that comprehensively covers various aspects to promote the value chain can boost investments. Nine of the 36 states and union territories, namely Andhra Pradesh, Bihar, Gujarat, Jharkhand, Karnataka, Maharashtra, Odisha, Tamil Nadu and Uttar Pradesh, have taken the lead by formulating dedicated electronics policies. Another 15 states and union territories have included electronics within the scope of their IT/ITeS policies.

A. Exports

Export target: Indian electronics exports stood at Rs 1,09,797 crore as of fiscal 2022, registering a growth of ~22% CAGR in the previous five years. Exports are further estimated to touch ~Rs 8,50,800 crore by fiscal 2026 (\$120 billion) as per Ministry of Electronics and Information Technology. While all the states provide export promotion, 3 states have declared exports target. [Gujarat](#) has set a manufacturing and export target of \$30 billion followed by [Maharashtra](#) of \$2 billion. [Tamil Nadu](#) has

set a target to contribute nearly 25% of India's total electronics exports by 2025. The PLI schemes are expected to be the drivers of electronics exports from India with Mobile phones at the forefront of it accounting for ~74% (Rs 6,50,000 crore) of exports. As per the Commerce and Industries Ministry, Electronics exports in fiscal 2023 recorded a 50.5% on-year growth, rising to \$23.57 billion from \$15.66 billion. Exports of mobile phones doubled to Rs 68,605 crore in 11 months of fiscal 2023 from 35,696 crore in 2022. Focus on mobile exports also reflects in policies as 7 out of thirty-six states and union territories have offered/covered incentives on Mobile manufacturing.

Note: USD to Rs conversion rate at Rs 70.9 (average rate of fiscal 2022)

B. Employment

Employment: Several states have taken various employment benefit initiatives ranging from creating incremental employment to training and developing workforce. Several Indian states offer incremental employment targets, incentivise development centres, promote employment of women and differently abled. [Andhra Pradesh](#) aims to establish 30 skill development centres and an IT skills university to cultivate a talent pool for electronics manufacturing. [Assam](#) plans to create an ICT and electronics fund while offering incentives to differently abled employees. [Haryana](#) and [Jharkhand](#) permit three-shift operations and women working at night with safety precautions.

C. Financial incentives

Electricity cost incentive: Power cost accounts for a significant share of the total cost of manufacturing companies after raw material cost. Several Indian states have designed unique power incentives to attract and foster electronics manufacturing. [Chhattisgarh](#) grants extended electricity duty exemption, while [Goa](#) offers waivers, quality power, and reimbursement for backup power and solar plant setup. [Gujarat](#) introduced power tariff subsidies and electricity duty exemption, [Uttar Pradesh](#) implements electricity duty exemptions based on regions. These comprehensive policies aim to nurture electronics manufacturing and growth in diverse regions across the country.

Capital incentives: Multiple Indian states have introduced comprehensive incentives to bolster the electronics manufacturing sector. Notably among them are [Andhra Pradesh's](#) capital support of up to 25% of Fixed Capital Investment (FCI), with an additional 20% for SC/ST/BC/Women enterprises. [UP](#) provides 50% reimbursement of fixed capital investment. [Haryana](#) tops up DEITY's support by 50% for units in designated Electronic Manufacturing Clusters. [Maharashtra](#) includes ESDM units under the Package Scheme of Incentives. [Tamil Nadu](#) and [Uttar Pradesh](#) also offer varying capital subsidies based on investment size and employment generation. These measures collectively aim to invigorate the electronics manufacturing landscape across the states

Interest incentives: As many as 15 out of thirty-six states and union territories in India offer interest incentives in various forms. Notable among them are [Chhattisgarh](#) reimbursing 75% of annual interest paid for 8 years. [Gujarat](#) provides interest assistance of 7% or actual interest paid, up to Rs 10 crore per year for 5 years. [Himachal Pradesh](#) gives a 5% interest subvention on term loans for 5 years. [Madhya Pradesh](#) provides 5% interest subsidy on loans up to Rs 10 crore and 2% on loans beyond, up to Rs 3 crore annually for 5 years.

Stamp duty and registration benefits: Waiver of stamp duty and registration fee is one of the incentives that states offer in various forms to improve ease of doing business. Andhra Pradesh offers full reimbursement of stamp duty, transfer duty and registration fees for the first sale/lease transaction. Gujarat offers one-time 100% stamp duty reimbursement for leases. Himachal Pradesh, Jharkhand and Madhya Pradesh also provide 100% reimbursement for stamp duty, transfer duty and registration fees. Karnataka exempts registered companies from stamp duty on loan agreements, lease deeds, and sale deeds for industrial plots, among others. Maharashtra exempts ESDM units from stamp duty during the investment period. Tamil Nadu provides stamp duty exemptions for land acquisition based on districts, and Uttar Pradesh offers various stamp duty exemptions for individual ESDM units and EMCs/ESDM parks.

Waivers of stamp duty & registration benefits: Waiver of stamp duty & registration fee is one of the factors that states offer in various forms to improve ease of doing business. [Andhra Pradesh](#) offers full reimbursement of stamp duty, transfer duty, and registration fees for the first sale/lease transaction. [Himachal Pradesh](#), [Jharkhand](#), and [Madhya Pradesh](#) also provide 100% reimbursement for stamp duty, transfer duty, and registration fees

D. Ease of doing business

Creation of land banks/incentives: Several Indian states are implementing strategies to encourage electronics manufacturing through land and infrastructure incentives. [Goa](#) provides land at subsidized rates and additional 20% rebate for ESDM units with capital investments within 3 years. [Haryana](#) permits anchor units to sublease 10% of their land to support service entities. [Tamil Nadu](#) provides land at subsidized rates and lease rental subsidies for eligible projects. [Uttar Pradesh](#) grants land subsidies based on region and land purchase, with stipulations on usage and FAR. These initiatives aim to enhance electronics manufacturing ecosystems across the country.

Other fiscal support: Other fiscal support includes logistics subsidies, tax rebates, market promotion incentives, and patent incentives. [Chhattisgarh](#) provides land premium and tax exemptions. [Goa](#) grants CST and entry tax reimbursements. [Gujarat](#) offers logistics and equipment cost subsidies. [Haryana](#) gives net tax reimbursement. [Maharashtra](#) and [Odisha](#) exempt property tax and offer SGST reimbursement. [Uttar Pradesh](#) aids patent filing and equipment import.

Single-window portal: While most states have a time-bound mechanism for clearances, 14 out of the 36 states and union territories offer a single-window portal to reduce the complexities and number of procedures in getting approval and licences. States have implemented e-platforms for simplicity, efficiency and transparency. These platforms include the provision for submitting combined applications and provide the turnaround time, among other things.

E. Other non-fiscal support

Other non-fiscal support: Other non-fiscal support includes self-certification, promotional and preferential market assistance, uninterrupted utility supply, and facility centres. [Assam](#) offers preferential market access with 50% domestically sourced Bill of Materials. [Gujarat](#) ensures regulatory approval and uninterrupted utility supply. [Haryana](#) strengthens IP Facilitation Centre to aid start-ups in intellectual property protection and commercialization. [Himachal Pradesh](#) and [Jharkhand](#) mandate local preference clauses for government purchases of software and hardware.

Incentivising R&D: Indian states focus on promoting Research and Development (R&D) in the electronics sector. [Haryana](#) plans to reimburse 50% of expenses for research scholars collaborating with global institutes. [Karnataka](#) encourages R&D projects in universities and colleges, fostering interdisciplinary centres and collaborations with top technical universities worldwide. [Odisha](#) offers reimbursement of 30% of R&D expenses and assistance for sponsored research work. [Uttar Pradesh](#) supports recognized R&D centres with financial reimbursement up to 50% of project costs over five years.

Incentivising EMCs: States in India prioritise Electronics Manufacturing Clusters (EMCs) to foster the growth of the electronics industry. As of June 2023, India had 80 operational EMCs spread across 22 states with a grant aid of Rs 1,470 crore. [Gujarat](#) and [Telangana](#) account of 39% of these grant aid and 31% of the total operational EMCs. Other states have specific promotions in place to attract investors and build EMCs. Some notables are [Andhra Pradesh's](#) focus on building new EMCs and sub-sector specific parks, while [Karnataka](#) plans to accelerate greenfield and brownfield cluster development. [Maharashtra](#) offers grant-in-aid for greenfield EMCs and assistance for brownfield EMCs, with contributions from the state and MIDC. [Uttar Pradesh](#) implements the Modified EMC 2.0 scheme, establishing EMCs and Common Facility Centres with financial support from the government and Project Implementing Agencies (PIAs).

Support for MSMEs: Micro, small and medium enterprises (MSMEs) are integral to the industrial ecosystem, and policies emphasize schemes to strengthen their global competitiveness through measures like interest and capital subsidies, market development assistance, and technology upgrades. [Chhattisgarh](#) offers reimbursements for ISO and patent certifications. [Haryana](#) treats vendor units equally to anchor units for incentives, including credit rating expense

reimbursement. [Maharashtra](#) provides incentives for MSMEs, including subsidies for technology up-gradation and cleaner production, as well as assistance for patent registration and quality certifications. These measures aim to enhance the growth of MSMEs by fostering innovation, quality improvement, and financial support.

E-waste management policy: E-waste management policy is crucial to prevent environmental harm, promote recycling, and ensure safe disposal of electronic waste. While majority states and union territories focus on e-waste management, few notable with incentives are [Assam's](#) policy of a minimum of 5 years of commercial operations for ESDM E-Waste Management units benefiting from the policy incentives. [Maharashtra](#) and [Uttar Pradesh](#) both aim to collaborate with industries to enhance e-waste management by restricting hazardous substance usage, fostering environmentally responsible practices in the electronics sector.

3. Glossary

EMCs: Electronic Manufacturing Clusters (EMCs) are specialized zones or areas designated by governments to foster the growth and development of the electronic manufacturing industry. These clusters aim to create an ecosystem that supports the entire electronic manufacturing value chain, from research and development to production and export. Typically equipped with state-of-the-art infrastructure, facilities, and regulatory support, EMCs attract and host electronics manufacturing units, component suppliers, research and development centers, and related enterprises. By concentrating resources and expertise in a geographically defined area, EMCs aim to enhance efficiency, innovation, and collaboration within the electronic manufacturing sector, contributing to economic growth and technological advancement.

E-waste: Electronic waste, commonly known as e-waste, refers to discarded electrical or electronic devices. These devices may include obsolete or broken household appliances, computers, mobile phones, televisions, and other electronic gadgets. E-waste is characterized by its potential environmental and health hazards due to the presence of toxic materials such as lead, mercury, cadmium, and various hazardous chemicals. Proper disposal and recycling of e-waste are critical to mitigate the environmental impact and prevent the release of harmful substances into the soil and water. E-waste management involves the collection, recycling, and safe disposal of electronic equipment to minimize environmental pollution and promote sustainable practices.

4. Table of Insight

																	
SI No.	Geography	Policy	Dedicated electronics policy	Export target	Employment	Electricity cost incentive	Capital incentive	Stamp duty and registration benefits	Interest incentive	Other fiscal support	Creation of land banks/ incentives	Single window portal	Other non-fiscal support	R&D incentive	EMCs incentive	Support for MSMEs	E-waste management policy
1	India	National Policy on Electronics 2019	✓	✓ Pg 20	✓ Pg 19		✓ Pg 17			✓ Pg 17				✓ Pg 24	✓ Pg 17	✓ Pg 23	✓ Pg 21
2	Andhra Pradesh	The Andhra Pradesh Electronics Policy, 2021	✓	✓ Pg 04	✓ Pg 02	✓ Pg 08	✓ Pg 07	✓ Pg 07	✓ Pg 08		✓ Pg 05	✓ Pg 07		✓ Pg 06	✓ Pg 05	✓ Pg 05	
3	Assam	Information Technology and Electronics Policy 2017			✓ Pg 19	✓ Pg 21	✓ Pg 20	✓ Pg 21		✓ Pg 22	✓ Pg 21	✓ Pg 23	✓ Pg 23				✓ Pg 14
4	Bihar	Bihar ESDM Vision 2017	✓		✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓	
5	Chandigarh	Information Technology & Electronics Policy, Chandigarh 2013												✓ Pg 10			
6	Chhattisgarh	Electronics, IT and ITES Investment Policy of Chhattisgarh 2015 - 2020			✓ Pg 13	✓ Pg 14	✓ Pg 12	✓ Pg 14	✓ Pg 12	✓ Pg 16	✓ Pg 13	✓ Pg 09	✓ Pg 16	✓ Pg 07	✓ Pg 09	✓ Pg 13	
7	Goa	Information Technology Investment Policy 2015-20			✓ Pg 05	✓ Pg 09		✓ Pg 08		✓ Pg 08	✓ Pg 08	✓ Pg 09					
8	Gujarat	Gujarat Electronics Policy 2022-2028 and Gujarat Semiconductor Policy 2022-27	✓	✓ Pg 03	✓ Pg 03	✓ Pg 08	✓ Pg 06	✓ Pg 07	✓ Pg 07	✓ Pg 06	✓ Pg 05	✓ Pg 13	✓ Pg 12	✓ Pg 12		✓ Pg 04	

																	
				Aspect: Exports	Aspect: Employment	Aspect: Financial incentives					Aspect: Ease of doing business		Aspect: Non fiscal support				
SI No.	Geography	Policy	Dedicated electronics policy	Export target	Employment	Electricity cost incentive	Capital incentive	Stamp duty and registration benefits	Interest incentive	Other fiscal support	Creation of land banks/ incentives	Single window portal	Other non- fiscal support	R&D incentive	EMCs incentive	Support for MSMEs	E-waste management policy
9	Haryana	Haryana Government Information Technology, Electronics Policy 2017			✓ Pg 30	✓ Pg 29	✓ Pg 26	✓ Pg 26		✓ Pg 28	✓ Pg 28	✓ Pg 27	✓ Pg 27	✓ Pg 30	✓ Pg 29	✓ Pg 30	✓ Pg 23
10	Himachal Pradesh	IT, ITeS and ESDM Policy - 2019 Himachal Pradesh			✓ Pg 07	✓ Pg 07	✓ Pg 09	✓ Pg 10	✓ Pg 10	✓ Pg 10	✓ Pg 11		✓ Pg 11				
11	Jharkhand	ESDM Policy 2016	✓		✓ Pg 13	✓ Pg 10	✓ Pg 10	✓ Pg 13	✓ Pg 21	✓ Pg 13	✓ Pg 13	✓ Pg 11	✓ Pg 10			✓ Pg 16	
12	Karnataka	Karnataka Electronics System Design & Manufacturing Policy 2017-2022	✓		✓ Pg 13	✓ Pg 29	✓ Pg 26	✓ Pg 28	✓ Pg 27	✓ Pg 27				✓ Pg 26	✓ Pg 16	✓ Pg 16	
13	Madhya Pradesh	Madhya Pradesh IT, ITeS & ESDM Investment Promotion Policy, 2016			✓ Pg 06	✓ Pg 12	✓ Pg 09	✓ Pg 10	✓ Pg 08	✓ Pg 12	✓ Pg 05	✓ Pg 04				✓ Pg 05	
14	Maharashtra	Maharashtra Electronics Policy, 2016	✓	✓ Pg 24	✓ Pg 25	✓ Pg 26	✓ Pg 25	✓ Pg 26	✓ Pg 25	✓ Pg 26		✓ Pg 30		✓ Pg 28	✓ Pg 27	✓ Pg 25	✓ Pg 25
15	Orissa	Odisha Electronics Policy 2021	✓		✓ Pg 07	✓ Pg 05	✓ Pg 05	✓ Pg 05	✓ Pg 08	✓ Pg 11	✓ Pg 05		✓ Pg 10	✓ Pg 09		✓ Pg 06	
16	Punjab	IT Policy-2013 with amendments			✓ Pg 05						✓ Pg 10						
17	Rajasthan	Rajasthan Investment Promotion Scheme 2019			✓ Pg 05	✓ Pg 05	✓ Pg 05	✓ Pg 05		✓ Pg 05	✓ Pg 05		✓ Pg 05		✓ Pg 02	✓ Pg 05	
18	Sikkim	Information Technology, Electronics and															✓ Pg 11

																	
SI No.	Geography	Policy	Dedicated electronics policy	Export target	Employment	Electricity cost incentive	Capital incentive	Stamp duty and registration benefits	Interest incentive	Other fiscal support	Creation of land banks/ incentives	Single window portal	Other non-fiscal support	R&D incentive	EMCs incentive	Support for MSMEs	E-waste management policy
		Telecommunication Policy 2021															
19	Tamil Nadu	Tamil Nadu Electronics Hardware Manufacturing Policy 2020	✓	✓ Pg 09	✓ Pg 13	✓ Pg 12	✓ Pg 12	✓ Pg 13	✓ Pg 12	✓ Pg 13	✓ Pg 13	✓ Pg 12	✓ Pg 13	✓ Pg 20	✓ Pg 15	✓ Pg 12	✓ Pg 19
20	Uttar Pradesh	Uttar Pradesh Electronics Manufacturing Policy 2020 (First Amendment-2022)	✓		✓ Pg 05	✓ Pg 11	✓ Pg 11	✓ Pg 11	✓ Pg 09	✓ Pg 11	✓ Pg 09	✓ Pg 05	✓ Pg 12	✓ Pg 08	✓ Pg 08	✓ Pg 08	✓ Pg 08
21	Uttarakhand	Information and Communication Technology & Electronics Policy 2016-2025			✓ Pg 07			✓ Pg 14		✓ Pg 12			✓ Pg 13				
22	West Bengal	Information Technology and Electronics Policy 2018					✓ Pg 05			✓ Pg 05					✓ Pg 05	✓ Pg 05	✓ Pg 05

5. Recommended Readings

PIB documents

1. Release ID: 1855316
2. Release ID: 1920586
3. MEITY Annual Report 2023
4. IT/ITeS and ESDM state policies
5. EMCPMS dashboard



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